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NON-TRANSITORY COMPUTER READABLE STORAGE MEDIUM AND INFORMATION PROCESSING APPARATUS

Abstract

A non-transitory computer readable storage medium stores a program that is a resident program. The program, when executed by a processor of an information processing apparatus, causes the information processing apparatus to perform operations including: acquiring a status of the image processing device through a communication with an image processing device to be monitored; displaying a first notification image based on the status acquired in the acquiring the status of the image processing device; acquiring service-related information related to a service available to the image processing device to be monitored; and displaying a second notification image related to the service-related information acquired in the acquiring the service-related information. A display format of the second notification image is the same as a display format of the first notification image.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority from Japanese Patent Application No. 2024-034174 filed on Mar. 6, 2024. The entire content of the priority application is incorporated herein by reference.

BACKGROUND ART

[0002] The technical field disclosed in the present specification relates to a non-transitory computer readable storage medium storing a resident program, and an information processing apparatus, particularly, the resident program is configured to communicate with an image processing device.

[0003] As an example of a resident program that functions as a resident process, which is resident after an information processing apparatus is activated and continues processing, there is a program for communicating with an image processing device to be monitored. For example, a status monitor caused to function by a resident program periodically communicates with a printer, acquires a status of the printer, and generates a notification of information indicating the status as necessary.

DESCRIPTION

[0004] The above-described resident program causes the information processing apparatus to communicate with the image processing device to be monitored and notify a user of information based on a status of the image processing device to be monitored. However, there is various information related to the image processing device in addition to the information based on the status, and there may be a need to effectively generate notifications of various information.

[0005] In an aspect of the present disclosure, there is provided a non-transitory computer readable storage medium storing a program that is a resident program, the program, when executed by a processor of an information processing apparatus, causing the information processing apparatus to perform operations, the operations including: communicating, via a communication device of the image processing apparatus, with an image processing device to be monitored; acquiring a status of the image processing device through a communication with the image processing device; displaying a first notification image based on the status acquired in the acquiring the status of the image processing device; acquiring service-related information related to a service available to the image processing device to be monitored; and displaying a second notification image related to the service-related information acquired in the acquiring the service-related information, a display format of the second notification image being the same as a display format of the first notification image.

[0006] According to the technique disclosed in the present specification, the program allows the information processing apparatus to communicate with the image processing device to be monitored and notify the user of the information based on the status, and furthermore, to notify the user of service-related information related to the image processing device to be monitored in the same display format as that of the information based on the status. Consequently, various information related to the image processing device to be monitored can be effectively provided to the user.

[0007] An information processing apparatus configured to achieve the operations by the above program is also novel and useful.

Description

[0008] According to the technique disclosed in the present specification, a technique of effectively generating notifications of various information is achieved in a resident program for communicating with an image processing device.

[0009] FIG. 1 is a schematic diagram of an information processing system according to a first embodiment.

[0010] FIG. 2 is a diagram illustrating a display of an operation display unit in a state in which a first status information screen is displayed as a pop-up.

[0011] FIG. 3A is a diagram illustrating a second status information screen displayed as a pop-up on the display of the operation display unit; FIG. 3B is a diagram illustrating a third status information screen displayed as a pop-up on the display of the operation display unit; and FIG. 3C is a diagram illustrating an offering information screen displayed as a pop-up on the display of the operation display unit.

[0012] FIG. 4 is a sequence diagram illustrating an example of a procedure for display processing of status information, offering information, and a notification list, which is implemented by a device monitoring program of an information processing apparatus, including operations of a device and a server.

[0013] FIG. 5 is a diagram illustrating an example of a notification list screen.

[0014] FIG. 6 is a sequence diagram illustrating an example of a procedure for display processing of status information and offering information, which is implemented by a device monitoring program and a device control program of an information processing apparatus, including operations of a device and a server, according to a second embodiment.

[0015] FIG. 7 is a diagram illustrating an example of a display in a state in which a task box is displayed as a pop-up.

[0016] FIG. 8 is a diagram illustrating a display method and a priority of a first badge.

[0017] FIG. 9 is a flowchart illustrating an example of a procedure for badge display processing, which is implemented by a CPU of the information processing apparatus sequentially processing according to commands described in the device monitoring program.

[0018] FIG. 10 is a diagram illustrating an example of the display in a state in which an activation box is displayed as a pop-up.

[0019] FIG. 11 is a diagram illustrating an example of the display in a state in which a second home screen is displayed.

FIRST EMBODIMENT

[0020] Hereinafter, a specific embodiment of an information processing system according to the present embodiment will be described in detail with reference to the accompanying drawings. In the present embodiment, the present disclosure is applied to the information processing system including a printer and an information processing apparatus that can communicate with each other.

[0021] That is, as illustrated in FIG. 1, an information processing system **100** according to the present embodiment includes four devices **1A**, **1B**, **1C**, and **1D** and an information processing apparatus **200**, which are communicatively connected to each other. In the following description, when there is no need to distinguish between the devices **1A**, **1B**, **1C**, and **1D**, the subscripts for distinguishing A to D will be omitted and the devices will be described as a device **1**. The device is an example of an image processing device.

[0022] The device **1** includes a printer **1A** capable of printing on a medium to be printed, and an image reading device **1B** capable of reading an image recorded on a medium to be read. In the present embodiment, the printer **1A** is exemplified as an inkjet printer. As a modification, the printer **1A** may be a laser printer, a label printer, a copier, or a multi function device. The information processing apparatus **200** generates and edits image data to be printed by the printer **1A**, transmits a print execution instruction and the image data to the printer **1A**, and receives status information from the printer **1A**.

[0023] The information processing apparatus **200** transmits a reading execution instruction to the image reading device **1B**, receives read data from the image reading device **1B**, and receives status information from the image reading device **1B**. The image reading device **1B** may be a copier, a multi function device, or the like.

[0024] The information processing apparatus **200** is, for example, a smartphone, a personal computer, or a tablet computer. The number of information processing apparatus **200** constituting the information processing system **100** is not limited to one. A plurality of information processing apparatus may be provided.

[0025] As illustrated in FIG. **1**, in the present embodiment, the printer **1A** includes a controller **11**, a print unit **12**, an operation panel **13**, a network interface **14**, and a storage unit **15**. The controller **11** includes a CPU and memories such as a flash ROM and a RAM, and controls each component of the printer **1A**.

[0026] The controller **11** is a general term for hardware used to control the printer **1A**, such as the CPU. Specifically, the controller **11** may also include an application specific integrated circuit (ASIC) and the like.

[0027] The print unit **12** is configured to print an image on a medium to be printed based on the image data transmitted from the information processing apparatus **200**. In the present embodiment, an inkjet printing method is adopted as an image forming method of the print unit **12**. The print unit **12** includes a print head that prints an image on a medium. Further, the print unit **12** may also be referred to as a print engine. The image forming method of the print unit **12** may be a thermal printing method or an electrophotographic method.

[0028] The operation panel **13** is, for example, a touch panel, receives an input from a user, and displays information. The operation panel **13** may include various display lamps, buttons, and the like.

[0029] The network interface **14** is hardware for communicating with the information processing apparatus **200**.

[0030] The storage unit **15** of the printer **1A** stores a program for an embedded web server (hereinafter, referred to as “EWS” (abbreviation of embedded web server)) **16**. Further, the storage unit **15** has a data storage area **17**, and the data storage area **17** stores, for example, capability information on the printer **1A**, connection information for connecting to the printer **1A**, status information on the printer **1A**, various setting information set in the printer **1A**, consumable remaining amount information on the printer **1A**, printed sheet number information on the printer **1A**, and consumable support-related information.

[0031] For example, the information processing apparatus **200** can access various information stored in the data storage area **17** via the EWS **16**. The printer **1A** manages a print progress status for each transmission source of a job. In the present embodiment, the consumable is an ink cartridge that contains ink consumed in print processing of a printer, and the remaining amount information is information related to a remaining amount of ink contained in the ink cartridge.

[0032] The devices **1B** to **1D** also include an EWS similar to the EWS **16** of the printer **1A**, and a storage unit (both not illustrated).

[0033] In the present embodiment, as illustrated in FIG. **1**, the information processing apparatus **200** includes a desktop personal computer (PC) **2** and an input and output device **25**. The desktop personal computer (PC) **2** includes a CPU **21**, a ROM **22**, a RAM **23**, a non-volatile memory **24**, an interface **26**, and a network interface **27**. The ROM **22** stores various programs and the like for the information processing apparatus **200**. The RAM **23** is used as a work area when various processing is executed, or as a storage area for temporarily storing data. The non-volatile memory **24** is, for example, a hard disk drive (HDD), a solid state drive (SSD), a flash memory, or the like, and stores various programs and data.

[0034] The CPU **21** executes various processing according to programs loaded from the ROM **22** to the non-volatile memory **24**.

[0035] The input and output device **25** is connected to the interface **26**, and includes an input device and an output device. The input device includes a keyboard **40**, a mouse **41**, and the like. The output device includes a display **42**. The display **42** is, for example, a liquid crystal display or an organic EL display. The input and output device **25** may be a touch panel that can be operated by touch. The information processing apparatus **200** may be a notebook PC or a tablet PC.

[0036] The network interface **27** is hardware for communicating with each device **1**. The network interface **27** is an example of a communication unit.

[0037] In the information processing system **100** according to the present embodiment, the information processing apparatus **200** and each device **1** perform wireless communication conforming to a Wi-Fi (registered trademark) standard via a common access point **3**. That is, the network interfaces **14** and **27** are both wireless interfaces that enable wireless communication based on the Wi-Fi (registered trademark) standard. The network interfaces **14** and **27** may be wired interfaces such as wired local area network (LAN) interfaces.

[0038] The communication mode according to the present embodiment is an example, and is not limited to the above mode. For example, the information processing apparatus **200** and each device **1** may communicate with each other by wireless communication through a direct connection without going through the access point **3**, or by wired communication using an interface such as USB.

[0039] The access point **3** is connected to a server **5** provided by a manufacturer, a sales company, or the like of the device **1** via an Internet **4**. Therefore, the information processing apparatus **200** can download a manual or the like of the device **1** from the server **5** via the Internet **4**. In addition to the manual, offering information such as service information and bargain information related to a consumable of the device **1** can be downloaded from the server **5**.

[0040] Examples of a service related to a consumable include an automatic consumable delivery service and a subscription service. The automatic consumable delivery service is a service that automatically delivers a consumable to a user of the printer based on consumable remaining amount information. The subscription service is a service that charges a fixed amount for up to a predetermined number of printed sheets of the printer, and automatically delivers the consumable to the user of the printer based on the consumable remaining amount information.

[0041] As illustrated in FIG. **1**, the non-volatile memory **24** of the information processing apparatus **200** stores an operating system (OS) **31**, an application program (app) **32**, a device monitoring program **33**, and a device control program **34**. Further, the nonvolatile memory **24** includes a data storage unit **35**.

[0042] The OS **31** is, for example, Windows (registered trademark), macOS (registered trademark), Linux (registered trademark), iOS (registered trademark), or Android (registered trademark). In the present embodiment, the OS **31** is Windows.

[0043] The app **32** is, for example, an image editing program having a function of receiving a user instruction via the interface **26**, a function of displaying an image on the display **42**, and a function of editing and saving the image. The app **32** may be a document editing program, a spreadsheet program, or the like.

[0044] The device monitoring program **33** is a resident program, and is a program for implementing a function of periodically communicating with each device **1** via the network interface **27** and acquiring various status information and the like from each device **1**. Further, the device monitoring program **33** is a program for implementing a function of periodically communicating with the server **5** via the network interface **27** and acquiring various information from the server **5**.

[0045] The resident program is, for example, a program that is automatically activated when the OS is activated and performs predetermined processing in a background during activation of the OS. The device monitoring program **33** is an example of a first program.

[0046] Specifically, the device monitoring program **33** acquires, from the printer **1A** via the

network interface 27, the capability information on the printer 1A, the connection information for connecting to the printer 1A, the status information on the printer 1A, the various setting information set in the printer 1A, the consumable remaining amount information on the printer 1A, the printed sheet number information on the printer 1A, the consumable support-related information, information on the print progress status, and the like, and stores the acquired information in the data storage unit 35.

[0047] Further, the device monitoring program 33 acquires, from the image reading device 1B via the network interface 27, capability information on the image reading device 1B, connection information for connecting to the image reading device 1B, status information on the image reading device 1B, various setting information set in the image reading device 1B, information related to a read progress status, and the like, and stores the acquired information in the data storage unit 35.

[0048] Further, the device monitoring program 33 acquires, via the network interface 27, the offering information related to the device 1 provided by the manufacturer or the sales company of the device 1, and stores the acquired offering information in the data storage unit 35. The offering information includes consumable-related service information and bargain information, maintenance information, notification of a campaign, a latest model, and introductions of convenient functions. Further, the offering information may include link information to the server 5. The offering information is an example of service-related information.

[0049] Then, the device monitoring program 33 can read the status information, the offering information, and the like of the device 1 stored in the data storage unit 35, and display the read information as a pop-up on the display 42.

[0050] The device control program 34 has a function of causing the printer 1A, which is set in a model name display field 73 described later, to print an image edited or stored by the app 32. That is, the device control program 34 receives a selection of an image to be printed, and transmits print data based on the selected image to the printer 1A, thereby causing the printer 1A to perform printing based on the print data. This function is an example of transmission processing. The status information, the consumable remaining amount information, and the like on the selected printer 1A can be read from the data storage unit 35 and displayed on the display 42. The device control program 34 is, for example, a non-resident program that is activated when activated and operated by the user. The device control program 34 is an example of a display program.

[0051] The device control program 34 instructs the selected image reading device 1B to read, receives image information read by the selected image reading device 1B, and stores the image information in the data storage unit 35. This function is an example of transmission processing. Further, information related to a read progress status of the selected image reading device 1B can be read from the data storage unit 35 and then displayed on the display 42.

[0052] The data storage unit 35 stores model names, model numbers, and the like of all the devices 1 to be monitored by the device monitoring program 33.

[0053] FIG. 2 is a diagram illustrating an example of a display state of the display 42 in a state in which a first status information screen 50 by the device monitoring program 33 is displayed as a pop-up. That is, on a first home screen 42A of the information processing apparatus 200, an icon 43 for the user to activate the device control program 34 is displayed on a left side of the display 42, and a taskbar 44 is displayed on a lower side of the display 42. A box display button 45 is displayed on a right side of the taskbar 44. The icon 43 for the device control program 34 is not required to be displayed on the first home screen 42A, and the device control program 34 may be activated by an operation on a menu screen of the information processing apparatus 200 (not illustrated).

[0054] The first home screen 42A is displayed on the display 42 by the OS, and on the first home screen 42A, the icon 43, the taskbar 44, the box display button 45, and the like are displayed by the OS.

[0055] On the first home screen 42A displayed on the display 42, a first status information screen 50 related to a consumable, for example, an ink remaining amount, is displayed as a pop-up. The

first status information screen **50** includes a first character string display field **50A** in which a character string such as “replace ink soon” is displayed as status information, a second character string display field **50B** in which a character string such as “access website containing support information” is displayed and which is linked to an access destination of the server **5**, a model name display field **50C** in which a model name of the device **1** is displayed, an ink remaining amount display field **50D** in which an ink remaining amount is displayed, a sheet remaining amount display field **50E** in which a sheet remaining amount is displayed, an offering information display field **50F** in which offering information or the like related to the status information displayed in the first character string display field **50A** is displayed, a notification list button display field **50G** in which a notification list button for displaying a notification list screen is displayed, and the like.

[0056] The model name display field **50C** displays a model name and an IP address of a corresponding device. Instead of the IP address, other identification information that can uniquely identify the device may be used. As other identification information, for example, a MAC address, a serial number, or the like can be used.

[0057] Further, on the first home screen **42A**, a second status information screen **51** not related to a consumable, for example, ink, as illustrated in FIG. **3A** is displayed as a pop-up by the device monitoring program **33**. The second status information screen **51** includes a model name display field **51A** in which the model name of the device **1** is displayed, and a second character string display field **51B** in which a character string such as “no sheet” is displayed as status information. When the character string “no sheet” is displayed in the second character string display field **51B**, the information processing apparatus **200** acquires status information indicating that there is no sheet from the printer **1**. This processing is an example of third display processing.

[0058] Further, on the first home screen **42A**, a third status information screen **52** not related to a consumable, for example, ink, as illustrated in FIG. **3B** is displayed as a pop-up by the device monitoring program **33**. The third status information screen **52** includes a model name display field **52A** in which the model name of the device **1** is displayed, and a progress status display field **52B** in which, for example, a scan progress status of the image reading device **1B** is displayed as status information.

[0059] Further, on the first home screen **42A**, an offering information screen **53** that displays offering information or the like as illustrated in FIG. **3C**, is displayed as a pop-up by the device monitoring program **33**. The offering information screen **53** includes an offering information display field **53A** in which offering information is displayed, a notification list button display field **53B** in which a notification list button for displaying a notification list screen is displayed, and the like.

[0060] The information processing apparatus **200** may be configured such that on the first home screen **42A**, the first status information screen **50** that displays the status information related to the consumable, the second status information screen **51** and the third status information screen **52** that display the status information not related to the consumable, and the offering information screen **53** that displays the offering information are displayed as pop-ups not to overlap each other, or these information screens **50** to **52** are overlapped and displayed as pop-ups.

[0061] Further, when the information processing apparatus **200** is configured such that a plurality of screens are overlapped and displayed as pop-ups, the information processing apparatus **200** may be configured such that a screen that displays new information is displayed as a pop-up in a foreground. In this case, since the screen that displays the new information is displayed as a pop-up in the foreground, the user can always keep an eye on a latest status of the device and latest information on the device, and acquire the status and the information, enabling a quick response.

Display of Status Information and Offering Information

[0062] FIG. **4** is a sequence diagram illustrating an example of a procedure for displaying the status information, the offering information, and the notification list, which is implemented by the device

monitoring program **33** of the information processing apparatus **200**, including operations of the device **1** and the server **5**.

[0063] Each processing step in the present embodiment represents processing of the CPU **21** according to an instruction described in a program such as the device monitoring program **33** of the information processing apparatus **200**. The processing by the CPU **21** also includes a hardware control using an API of the OS. In the present specification, a detail description of the OS is omitted, and an operation of each program is described. In addition, in the present specification, for convenience, processing primarily performed by the CPU **21** may be described as being primarily performed by a program or a device.

Display of Status Information and Offering Information

[0064] In FIG. **4**, first, in a case in which an error occurs during operation, for example, in a case in which the device **1** is the printer **1A**, when a consumable such as ink runs out or an error such as a sheet jam occurs during printing, the device **1** transmits status information indicating error content to the information processing apparatus **200** (procedure **11** (hereinafter, referred to as **T11**)). Therefore, the device monitoring program **33** that receives the status information stores the received status information in the data storage unit **35** (**T12**). This procedure is an example of first acquisition processing and fourth acquisition processing.

[0065] The device monitoring program **33** queries each device **1** connected to the information processing apparatus **200**, that is, each device **1** to be monitored by the device monitoring program **33** for the status information. In other words, the information processing apparatus **200** queries each device **1** at regular intervals, performing a so-called polling (**T13**). A polling interval for each device **1** may be constant, or may be varied based on, for example, a frequency of use of the device **1**.

[0066] Each device **1** that receives the query from the information processing apparatus **200** transmits the status information to the information processing apparatus **200** (**T14**). Therefore, the device monitoring program **33** that receives the status information stores the received status information in the data storage unit **35** (**T15**). This procedure is an example of the first acquisition processing, third acquisition processing, and the fourth acquisition processing.

[0067] When the status information is stored in the data storage unit **35** in **T12** or **T15**, the device monitoring program **33** determines whether the stored status information is status information related to a consumable such as ink provided by the manufacturer or the sales company of the device **1**. When the information processing apparatus **200** determines that the status information is status information related to the consumable, as illustrated in FIG. **2**, the device monitoring program **33** displays the first status information screen **50** as a pop-up on the first home screen **42A** based on content of the status information (**T16**). This processing is an example of first display processing.

[0068] Further, the device monitoring program **33** searches for offering information stored in the data storage unit **35** based on the content of the status information and the model name of the device **1**, specifically, for example, warning information “replace ink soon” displayed in the first character string display field **50A** and a model name “DCP-XXXDW” displayed in the model name display field **50C**, and extracts the offering information stored in association with the warning information and the model name. Then, the information processing apparatus **200** displays the extracted offering information in the offering information display field **50F** on the first status information screen **50**. This processing is an example of second display processing.

[0069] As described above, in the present embodiment, on the first status information screen **50** based on the status information related to the consumable among the status information transmitted from all the devices **1** connected to the information processing apparatus **200**, that is, the respective devices **1** to be monitored by the device monitoring program **33**, the offering information related to the warning information and the model name is also displayed in the offering information display field **50F**, and therefore, by viewing the content displayed as a pop-up, the user can recognize that

there is a device **1** having a predetermined status, or that there is offering information related to the status, thereby improving user convenience. Further, an improvement in user ability to cope with the status information can also be expected.

[0070] At this time, as the offering information displayed in the offering information display field **50F**, only the offering information unread by the user may be displayed, or all the offering information related to the warning information and the model name may be displayed regardless of whether the offering information is read or unread by the user. The offering information is stored in the data storage unit **35** in association with information indicating whether the offering information is unread or read. Information indicating the offering information is unread is associated immediately after being received from the server **5**, and when the offering information is displayed on the first status information screen **50** or the like, information indicating the offering information is read is associated instead of the information indicating the offering information is unread.

[0071] In the present embodiment, the device monitoring program **33** is configured to display the offering information stored in the data storage unit **35** in the offering information display field **50F** on the first status information screen **50**. However, the device monitoring program **33** may be configured to download the offering information from the server **5** by querying the server **5** for offering information based on the content of the status information, the model name of the device **1**, and the like, and display the offering information in the offering information display field **50F** on the first status information screen **50** based on the offering information downloaded from the server **5**.

[0072] Next, the device monitoring program **33** determines whether the status information stored in the data storage unit **35** is status information related to a consumable such as ink provided by the manufacturer or the sales company of the device **1**. When the device monitoring program **33** determines that the status information is not status information related to the consumable provided by the manufacturer or the sales company of the device **1**, such as running out of print sheet, the device monitoring program **33** displays the second status information screen **51** as illustrated in FIG. **3A** or the third status information screen **52** as illustrated in FIG. **3B** as pop-ups on the first home screen **42A** based on the content of the status information (**T17**). The second status information screen **51** is a screen indicating that there is no sheet in the device, and the third status information screen **52** is a screen indicating that scan processing is being executed in the device. This processing is an example of the first display processing and the third display processing, and the second status information screen **51** is an example of a third notification screen.

[0073] As described above, in the present embodiment, in a case in which the status information is not status information related to a consumable such as ink provided by the manufacturer or the sales company of the device **1**, that is, for example, no offering information is displayed on the second status information screen **51** that notifies the user of out-of-sheet or the third status information screen **52** that displays the scan progress status of the image reading device **1B**, and therefore, no unnecessary information is provided to the user, improving visibility of the user.

[0074] When the device monitoring program **33** receives the same status information from the device **1** while displaying a screen based on certain status information from the device **1** as a pop-up, the device monitoring program **33** may omit further displaying a screen based on the received status information as a pop-up.

[0075] Further, the device monitoring program **33** queries the server **5** connected to the information processing apparatus **200** for offering information, that is, the information processing apparatus **200** performs a so-called polling on the server **5** (**T18**). The query is accompanied by information indicating a time of a previous query. When offering information not transmitted to the information processing apparatus **200** is present, that is, new offering information registered after a time indicated by the accompanied information is present, the server **5** that receives the query transmits the new offering information to the information processing apparatus **200** (**T19**). Therefore, the device monitoring program **33** that receives the offering information stores the received offering

information in the data storage unit **35** that is accessible by the device control program **34** (T20).

This procedure is an example of second acquisition processing.

[0076] A polling interval for the server **5** may be the same as the polling interval for each device **1** performed by the information processing apparatus **200**, or the polling intervals for the device **1** and the server **5** may be different. For example, the polling interval for the server **5** may be set longer than the polling interval for the device **1**. Further, the polling interval for the server **5** may be changed based on circumstances such as a season or presence or absence of an event.

[0077] Next, the device monitoring program **33** searches for offering information stored in the data storage unit **35**, and when newly arrived offering information is present, the device monitoring program **33** displays the offering information screen **53** as illustrated in FIG. 3C as a pop-up based on the newly arrived offering information (T21). That is, similarly to the first status information screen **50** related to the status information, the device monitoring program **33** displays the offering information screen **53** related to the offering information as a pop-up. This processing is an example of the second display processing, and the offering information screen **53** is an example of a second notification image.

[0078] In this processing, when no newly arrived offering information is present, the device monitoring program **33** may display the offering information screen **53** as a pop-up on the first home screen **42A** based on latest offering information. Further, when no newly arrived offering information is present and the first status information screen **50** is displayed on the first home screen **42A**, the device monitoring program **33** may search for offering information stored in the data storage unit **35** based on the model name displayed in the model name display field **50C** on the first status information screen **50**, and may display the offering information screen **53** as a pop-up on the first home screen **42A** based on the extracted offering information.

Display of Notification List

[0079] Next, in a state in which the information processing apparatus **200** displays the status information screen **50** or displays the offering information screen **53**, when the user clicks a notification list button portion displayed in the notification list button display field **50G** on the first status information screen **50** or the notification list button display field **53B** on the offering information screen **53** with the mouse **41** (T31), the device monitoring program **33** displays all notifications for the monitored device **1** on the display **42** as a notification list screen **54**, based on the notification information stored in the data storage unit **35** (T32).

[0080] The notification list screen **54** displays notification objects **54A** to **54D**. Further, each of the notification objects **54A** to **54D** displays a notification date, a notification message, a target model name, and the like. Then, when a click operation with the mouse **41** on each of the objects **54A** to **54D** is received, a detailed information screen or a reference information screen are displayed.

[0081] The device control program **34** refers to the offering information stored in the data storage unit **35** by the device monitoring program **33** in T24, and displays the notification list screen **54** based on the referred offering information. Even when the user clicks a “notification list” tab **83A** portion with the mouse **41**, the notification list screen **54** can be displayed on a setting screen **80**.

[0082] As described above, in the present embodiment, since the notification information for all the devices **1** monitored by the device monitoring program **33** is stored in the data storage unit **35**, and the device monitoring program **33** displays the notification list screen **54** on the display **42** based on the notification information stored in the data storage unit **35**, the user can properly understand the offering information currently provided to the device **1**.

[0083] Further, in a state in which the device monitoring program **33** displays the first status information screen **50** or the offering information screen **53**, when the user clicks a character portion or an illustration portion on the display screen of the first status information screen **50** or the offering information screen **53** with the mouse **41** (T33), the device monitoring program **33** activates the device control program **34**, and as illustrated in FIG. 5, displays a detailed information screen or a reference information screen (not illustrated) on the display **42** for all notifications for

the monitored device **1**, based on the notification information stored in the data storage unit **35** (T34).

[0084] In the present embodiment, the information processing apparatus **200** is configured, in response to a display instruction, to display the notification list screen **54**, the detailed information screen, the reference information screen, and the like on the display **42** based on the notification information stored in the data storage unit **35**. However, the information processing apparatus **200** may also be configured, in response to a display instruction (T31, T33), to access the server **5** and display the notification list screen **54** and the like on the display **42** based on latest notification information downloaded therefrom (T32, T34). With this configuration, the user can view the latest notification list screen **54** and the like related to the device **1**, and can prevent an erroneous operation based on old information.

[0085] In the present embodiment, the device monitoring program **33** is configured to display the notification list screen **54**, the detailed information screen, and the reference information screen. However, the device monitoring program **33** may be configured to activate the device control program **34** to display the notification list screen **54**, the detailed information screen, the reference information screen, and the like.

[0086] As described above, in the present embodiment, since the notification list screen **54**, the detailed information, and the reference information are displayed on the display **42** based on an operation of the user, the information desired by the user can be provided in a timely manner, improving convenience. Further, since the detailed information and the reference information are acquired from the server **5**, the user can always acquire the latest information, improving convenience.

[0087] The notification information displayed on the notification list screen **54**, the detailed information and reference information screens displayed on the display **42** may be configured to display only information unread by the user on the display **42**, or may be configured to display information limited by an acquisition period, an issuance period, or the like on the display **42**. Further, it may be configured to preferentially display information related to a campaign or the like.

SECOND EMBODIMENT

[0088] Next, FIG. **6** is a sequence diagram illustrating an example of a procedure for display processing of status information and offering information, which is implemented by the device monitoring program **33** and the device control program **34** of the information processing apparatus **200**, including operations of the device **1** and the server **5**.

[0089] Each processing procedure in the present embodiment represents processing of the CPU **21** according to an instruction described in programs such as the device monitoring program **33** and the device control program **34** of the information processing apparatus **200**. The processing by the CPU **21** also includes a hardware control using an API of the OS. In the present specification, a detail description of the OS is omitted, and an operation of each program is described. In the description, the same components as those in the first embodiment are denoted by the same reference numerals. In addition, in the present specification, for convenience, processing primarily performed by the CPU **21** may be described as being primarily performed by a program or a device.

Display of Status Information and Offering Information

[0090] That is, when the user clicks a box display button **45** portion displayed in a notification area of the taskbar **44** displayed on the display **42** with the mouse **41** to instruct a display of a box (T41), as illustrated in FIG. **7**, the device monitoring program **33** displays a box **60** on the display **42** by the OS **31**, and displays, within the box **60**, a first icon **61** of the device monitoring program **33**, which is a resident program (T42). The first icon **61** is displayed accompanied by a first badge **62**. The box **60** is an area in which icons of resident programs can be displayed, and can also be referred to as an icon display area.

[0091] As illustrated in FIG. **7**, the device monitoring program **33** changes a display method of the

first badge **62** based on a type of information acquired from the monitored device, for example, the device **1** or the server **5** via the network interface **27**.

[0092] Specifically, when the status information acquired from each device **1** to be monitored includes status information related to an error, the device monitoring program **33** displays, for example, a red first badge **62A**. Further, when the status information acquired from each device **1** does not include the status information related to an error but includes status information related to a warning, the device monitoring program **33** displays, for example, a yellow first badge **62B**.

[0093] Further, when the status information acquired from each device **1** to be monitored does not include the status information related to the error and the information related to the warning, and new offering information is acquired from the server **5**, that is, when unread offering information is stored in the data storage unit **35**, the device monitoring program **33** displays, for example, a first badge **62C** with a white bell mark on a pink background.

[0094] Further, when the status information acquired from each device **1** to be monitored does not include the status information related to an error and the information related to a warning, does not store unread offering information, but includes status information related to readiness, the device monitoring program **33** displays, for example, a green first badge **62D**. When the information processing apparatus **200** does not store unread offering information and all the devices **1** to be monitored are offline, the information processing apparatus **200** displays, for example, a gray first badge **62E**. In the following description, when there is no need to distinguish the first badges **62A** to **62E**, the subscripts for distinguishing A to E will be omitted and the first badges **62A** to **62E** will be described as the first badge **62**. The first badge **62** is an example of an icon.

[0095] In a display of the first badge **62** in the box **60**, a priority is determined based on a type of the status information and the offering information, and the device monitoring program **33** displays one first badge **62** among the first badges **62A** to **62E** based on the priority.

[0096] Specifically, as illustrated in FIG. **8**, the first badge **62** having a highest display priority is the red first badge **62A** indicating that the status information is an error. The first badge **62** having a next highest priority is the yellow first badge **62B** indicating that the status information is a warning. The first badge **62** having a next highest priority is the first badge **62C** indicating that unread offering information is stored. The first badge **62** having a next highest priority is the green first badge **62D** indicating that the status information is ready. The first badge **62** having a next highest priority is the gray first badge **62E** indicating that the status information is offline.

[0097] As described above, in the present embodiment, since the badge is displayed regardless of whether the notification is related to the status information or the offering information, the user can understand what information is being notified by viewing the badge. In addition, in a case in which the status information is an error or a warning, since the display priority is set to be higher than that of other status information, the user can quickly respond to the error or the warning.

[0098] The red first badge **62A** and the yellow first badge **62B** each are an example of a first mode, and the first badge **62C** is an example of a second mode.

[0099] FIG. **9** is a flowchart illustrating an example of a procedure for badge display processing, which is implemented by the CPU **21** of the information processing apparatus **200** sequentially processing according to commands described in the device monitoring program **33**. The badge display processing is periodically executed by the CPU **21** at regular intervals. Alternatively, the processing may be executed every time status information or offering information is acquired.

[0100] First, the information processing apparatus **200** reads status information and offering information stored in the data storage unit **35** (step **51** (hereinafter, referred to as **S51**)).

[0101] Next, the device monitoring program **33** determines whether new status information or unread offering information is present (**S52**). The “new status information” here means that the status of the device **1** to be monitored has changed. If the information processing apparatus **200** determines that no new status information or unread offering information is present (**S52**: NO), the device monitoring program **33** maintains the current display of the first badge **62** (**S53**), and this

processing ends.

[0102] If the device monitoring program **33** determines that new status information or unread offering information is present (S52: YES), then the device monitoring program **33** determines whether the first badge **62** is currently displayed (S54). If the device monitoring program **33** determines that the first badge **62** is not currently displayed (S54: NO), then the device monitoring program **33** displays the first badge **62** determined based on the priority described above based on the new status information or the unread offering information (S55), and then this processing ends.

[0103] If the device monitoring program **33** determines that the first badge **62** is currently displayed (S54: YES), the device monitoring program **33** determines whether the new status information or the unread offering information has a higher priority than the first badge **62** currently displayed (S56). If the device monitoring program **33** determines that the new status information or the unread offering information has a higher priority than the currently displayed first badge **62** (S56: YES), then the device monitoring program **33** proceeds to processing in S55, and displays the first badge **62** corresponding to the new status information or the unread offering information, and then this processing ends.

[0104] If the device monitoring program **33** determines that the new status information or the unread offering information does not have a higher priority than the currently displayed first badge **62** (S56: NO), then the information processing apparatus **200** proceeds to processing in S53, and maintains the current display of the first badge **62**, and this processing ends.

[0105] The determination of the priority is executed based on the priority illustrated in FIG. **8** described above. In the present embodiment, the display priority of the first badge **62C** indicating that the unread offering information is acquired is set to be higher than the green first badge **62D** indicating that the status information is ready and the gray first badge **62E** indicating that the status information is offline, and is set to be lower than the red first badge **62A** indicating that the status information is an error and the yellow first badge **62B** indicating that the status information is a warning. However, the display priority of the first badge **62C** is not limited thereto. Further, the priority of the status information may be set based on a type of the unread offering information.

[0106] Returning to the sequence diagram illustrating the example of the procedure for the display processing of the status information and the offering information illustrated in FIG. **6**, when the user clicks a first icon **61** portion illustrated in FIG. **7** with the mouse **41** to instruct a display of an activation box **63** (T43), the information processing apparatus **200** displays the activation box **63** on the display **42**, as illustrated in FIG. **10** (T44).

[0107] In the activation box **63**, characters **63A** “activate control program”, characters **63B** “notification list”, and the like are displayed, respectively.

[0108] Here, when the user clicks a portion of the characters **63A** “activate control program” in the activation box **63** with the mouse **41** to instruct activation of the device control program **34** (T45), the device monitoring program **33** activates the device control program **34** (T46).

[0109] The device control program **34** can also be activated by the user clicking an icon **43** portion illustrated in FIG. **2** with the mouse **41**. Further, a shortcut icon for activating the device control program **34** may be provided in the taskbar **44**. When the user clicks a portion of the characters **63B** “notification list” with the mouse **41**, the device monitoring program **33** instructs the device control program **34** to display the notification list screen **54**.

[0110] Here, when the device control program **34** is activated in the device monitoring program **33**, the device control program **34** displays a second home screen **70** of the device control program **34** on the display **42** (T47).

[0111] As illustrated in FIG. **11**, on the second home screen **70**, icons corresponding to functions available in the selected printer, for example, a print icon **71** and a scan icon **72** are displayed. The second home screen **70** includes a control model name display field **73** indicating a name of the selected printer, a status information display field **74** indicating a status of the selected printer, and an ink remaining amount display field **75** indicating a consumable remaining amount of the

selected printer.

[0112] That is, the print icon **71** is used to instruct the selected printer to perform printing. The scan icon **72** is used to instruct the selected printer to read an image. In the control model name display field **73**, one of the devices **1** to be monitored by the device monitoring program **33** is selected as a control target of the device control program **34**, and a name of the selected device, specifically, a model name of the device **1** is displayed. Further, the status information on the device **1** displayed in the control model name display field **73** is displayed in the status information display field **74**. When the device **1** displayed in the control model name display field **73** is the printer **1A**, an ink remaining amount of the printer **1A** is displayed in the ink remaining amount display field **75**.

[0113] When the device control program **34** is activated, the device control program **34** displays a second icon **76** of the device control program **34** in the taskbar **44** on the first home screen **42A** (**T48**). The second icon **76** is displayed accompanied by a second badge **77** as necessary.

Specifically, the device control program **34** displays, in the second badge **77**, the number of unread offering information among the offering information for all the devices **1** to be monitored by the device monitoring program **33**, which is stored in the data storage unit **35**. That is, the second badge **77** illustrated in FIG. **11** indicates a state in which 20 pieces of unread offering information related to all the devices **1** to be monitored by the device monitoring program **33** are present. When the number of pieces of unread offering information is 0, the second badge **77** may not be displayed, or the second badge **77** indicating 0 may be displayed.

[0114] As described above, in the present embodiment, when the device control program **34** is activated, the second icon **76** and the second badge **77** are displayed in the taskbar **44**, and thus the user can easily know the number of pieces of unread offering information related to all the devices **1** by the second badge **77**. Therefore, an opportunity for the user to read the offering information increases, and as a result, an opportunity to receive the service increases.

[0115] When status information on an error or status information on a warning having a higher priority than the unread offering information is acquired from the device **1** having a model name displayed in the control model name display field **73**, the red or yellow second badge **77** is displayed in the taskbar **44** on the second home screen **70** instead of the number of pieces of the unread offering information.

[0116] As described above, in the present embodiment, when the status information on an error or the status information on a warning having a higher priority than the unread offering information is acquired, the red or yellow second badge **77** is displayed in the taskbar **44** on the second home screen **70** instead of the number of pieces of the unread offering information, and thus the user can quickly respond to the error or the warning.

[0117] As described in detail above, according to the first embodiment and the second embodiment disclosed in the present specification, the device monitoring program **33** can effectively provide various information on the device to be monitored to the user by displaying a notification related to the offering information related to the device to be monitored in the same display format as the information based on the status, that is, displaying both notifications in a display format of a pop-up display. Further, the monitoring program can also effectively provide various information on the device to be monitored to the user by displaying a notification related to the offering information related to the device to be monitored in the same display format as the information based on the status, that is, displaying both notifications in a display format using icon badges. The display format using icon badges is an example of a display format via an icon.

[0118] The present embodiment is merely an example, and does not limit the present disclosure. Therefore, various improvements and modifications can be naturally made to the present disclosure without departing from the scope thereof. For example, the printer **1A** may not include the operation panel **13**.

[0119] The present embodiment is merely an example, and does not limit the present disclosure. Therefore, various improvements and modifications can be naturally made to the present disclosure

without departing from the scope thereof. For example, the printer **1A** may not include the operation panel **13**. The information processing apparatus **200** is not limited to the non-volatile memory **24**, and may include any type of large-capacity storage device.

[0120] In the present embodiment, a difference in the mode is expressed by changing display colors of the first badge and the second badge based on a type of the status information. However, the difference in the mode may be expressed by presence or absence of blinking of the first badge and the second badge or a change in blinking interval.

[0121] Further, in the processing in **T12**, when the information processing apparatus **200** receives highly urgent status information such as a sheet jam, the information processing apparatus **200** may be configured to immediately display any one of the first to third status information screens **50** to **52** on the display **42** based on the received status information to notify the user.

[0122] The communication method between the printer **1A** and the information processing apparatus **200** is not limited to wireless communication conforming to the Wi-Fi (registered trademark) standard. For example, the communication method may be wired communication using a USB cable or wireless communication based on other standards such as Bluetooth (registered trademark). Further, a plurality of communication functions may be provided.

[0123] The processing disclosed in the embodiment may be executed by hardware such as a single CPU, a plurality of CPUs, an ASIC, or a combination thereof. In addition, the processing disclosed in the embodiment can be implemented in various modes such as a recording medium in which a program for executing the processing is recorded, or a method.

Claims

1. A non-transitory computer readable storage medium storing a program that is a resident program, the program, when executed by a processor of an information processing apparatus, causing the information processing apparatus to perform operations, the operations comprising:

communicating, via a communication device of the image processing apparatus, with an image processing device to be monitored; acquiring a status of the image processing device through a communication with the image processing device; displaying a first notification image based on the status acquired in the acquiring the status of the image processing device; acquiring service-related information related to a service available to the image processing device to be monitored; and displaying a second notification image related to the service-related information acquired in the acquiring the service-related information, a display format of the second notification image being the same as a display format of the first notification image.

2. The non-transitory computer readable storage medium according to claim 1, wherein the operations comprise displaying an icon corresponding to the program, each of the display format of the first notification image and the display format of the second notification image being a display format via the icon, wherein in the displaying the first notification image, the icon in a first mode is displayed as the first notification image, the icon in the first mode corresponding to the status acquired in the acquiring the status of the image processing device, and wherein in the displaying the second notification image, the icon in a second mode is displayed as the second notification image, the icon in the second mode indicating that the service-related information acquired in the acquiring the service-related information is present.

3. The non-transitory computer readable storage medium according to claim 2, wherein the operations comprise, in a case in which the status acquired in the acquiring the status of the image processing device includes a content indicating an error or a warning, displaying the icon in the first mode while restricting a display of the icon in the second mode.

4. The non-transitory computer readable storage medium according to claim 3, wherein the operations comprise, in a case in which the status acquired in the acquiring the status of the image processing device does not include the content indicating the error or the warning and in a case in

which there is unread service-related information in the service-related information acquired in the acquiring the service-related information, displaying the icon in the second mode while restricting a display of the icon in the first mode.

5. The non-transitory computer readable storage medium according to claim 2, wherein the operations further comprise activating a display program that is a non-resident program, the display program, when executed by the processor, causing the information processing apparatus to perform display operations comprising displaying a list of the service-related information acquired in the acquiring the service-related information, the display operations further comprising, in response to activation of the display program, displaying an icon corresponding to the display program, the icon corresponding to the display program including a specific mode indicating that unread service-related information in the service-related information acquired in the acquiring the service-related information is present, and wherein the second mode of the icon corresponding to the program is different from the specific mode of the icon corresponding to the display program.

6. The non-transitory computer readable storage medium according to claim 5, wherein the icon in the specific mode indicates a number of pieces of the unread service-related information, and wherein the icon in the second mode does not indicate the number of pieces of the unread service-related information.

7. The non-transitory computer readable storage medium according to claim 2, wherein the icon in the first mode is an icon to which a first badge with a first color is assigned, the first color being determined in accordance with a content of the status from among one or more different colors for the first badge, and wherein the icon in the second mode is an icon to which a second badge with a second color is assigned, the second color being not included in the one or more different colors for the first badge.

8. The non-transitory computer readable storage medium according to claim 1, wherein each of the display format of the first notification image and the display format of the second notification image is a pop-up display, wherein in the displaying the first notification image, the first notification image is displayed as a pop-up, and wherein in the displaying the second notification image, the second notification image is displayed as a pop-up.

9. The non-transitory computer readable storage medium according to claim 8, wherein the operations further comprise: acquiring progress information indicating a progress of a job causing the image processing device to perform image processing; and displaying a third notification image based on the progress information acquired in the acquiring the progress information, a display format of the third notification image being the same as the display format of the first notification image.

10. The non-transitory computer readable storage medium according to claim 9, wherein the operations further comprise acquiring consumable information indicating information on a consumable in the image processing device through the communication with the image processing device, wherein the first notification image includes the consumable information acquired in the acquiring the consumable information, and wherein the third notification image does not include the consumable information.

11. The non-transitory computer readable storage medium according to claim 8, wherein the program, when executed by the processor, allows the information processing apparatus to monitor a plurality of image processing devices, wherein the displaying the first notification image comprises: receiving a selection of an image processing device to be displayed among the plurality of image processing devices to be monitored; and displaying the first notification image based on the status of the selected image processing device, and wherein the displaying the second notification image comprises displaying the second notification image without receiving the selection of the image processing device to be displayed.

12. The non-transitory computer readable storage medium according to claim 8, wherein the service-related information includes information related to one or more services associated with

respective one or more types of error or warning, wherein the operations further comprise, in a case in which the status acquired in the acquiring the status of the image processing device is an error or a warning, extracting, from the service-related information acquired in the acquiring the service-related information, information related to a service associated with a type of error or warning corresponding to the type of error or warning indicated in the status acquired in the acquiring the status of the image processing device, and wherein the displaying the first notification image comprises, in case in which there is the information extracted in the extracting the information related to the service, displaying the first notification image including the information extracted in the extracting information related to the service.

13. The non-transitory computer readable storage medium according to claim 8, wherein the acquiring the service-related information comprises storing the acquired service-related information in a memory of the information processing apparatus, and wherein the operations further comprise: receiving an instruction to display a list of the acquired service-related information during the second notification image being displayed; and displaying a list of the service-related information stored in the memory in a case in which the instruction is received.

14. The non-transitory computer readable storage medium according to claim 1, wherein the acquiring the service-related information comprises: communicating, via the communication device, with a server configured to provide the service-related information; and acquiring the service-related information related to the image processing device to be monitored.

15. An information processing apparatus comprising: a communication device; a processor; and a memory storing a program that is a resident program, the program, when executed by the processor, causing the information processing apparatus to perform operations, the operations comprising: communicating, via the communication device, with an image processing device to be monitored; acquiring a status of the image processing device through a communication with the image processing device; displaying a first notification image based on the status acquired in the acquiring the status of the image processing device; acquiring service-related information related to a service available to the image processing device to be monitored; and displaying a second notification image related to the service-related information acquired in the acquiring the service-related information, a display format of the second notification image being the same as a display format of the first notification image.

16. A method for controlling an information processing apparatus, the information processing apparatus comprising a processor and a communication device, the method comprising: executing a program by the processor, the program being a resident program, to cause the information processing apparatus to perform operations, the operations comprising: communicating, via the communication device, with an image processing device to be monitored; acquiring a status of the image processing device through a communication with the image processing device; displaying a first notification image based on the status acquired in the acquiring the status of the image processing device; acquiring service-related information related to a service available to the image processing device to be monitored; and displaying a second notification image related to the service-related information acquired in the acquiring the service-related information, a display format of the second notification image being the same as a display format of the first notification image.

17. The method according to claim 16, wherein the operations comprise displaying an icon corresponding to the program, each of the display format of the first notification image and the display format of the second notification image being a display format via the icon, wherein in the displaying the first notification image, the icon in a first mode is displayed as the first notification image, the icon in the first mode corresponding to the status acquired in the acquiring the status of the image processing device, and wherein in the displaying the second notification image, the icon in a second mode is displayed as the second notification image, the icon in the second mode indicating that the service-related information acquired in the acquiring the service-related

information is present.

18. The method according to claim 17, wherein the operations comprise, in a case in which the status acquired in the acquiring the status of the image processing device includes a content indicating an error or a warning, displaying the icon in the first mode while restricting a display of the icon in the second mode.

19. The method according to claim 18, wherein the operations comprise, in a case in which the status acquired in the acquiring the status of the image processing device does not include the content indicating the error or the warning and in a case in which there is unread service-related information in the service-related information acquired in the acquiring the service-related information, displaying the icon in the second mode while restricting a display of the icon in the first mode.

20. The method according to claim 17, wherein the operations further comprise activating a display program that is a non-resident program, the display program, when executed by the processor, causing the information processing apparatus to perform display operations comprising displaying a list of the service-related information acquired in the acquiring the service-related information, the display operations further comprising, in response to activation of the display program, displaying an icon corresponding to the display program, the icon corresponding to the display program including a specific mode indicating that unread service-related information in the service-related information acquired in the acquiring the service-related information is present, and wherein the second mode of the icon corresponding to the program is different from the specific mode of the icon corresponding to the display program.
